SPHOORTHY ENGINEERING COLLEGE | *(UGC Autonomous — NAAC Grade A)*

VLSI Design: CMOS Technology, Stick Diagrams, Layout & Verilog HDL

Department: ECE	**Course / Subject:** VLSI Design (EC601PC)
Semester: Semester 6	**Regulation:** Autonomous (SR23 / SR24)

Course Overview & Key Focus: MOS transistor fabrication, CMOS inverter DC transfer characteristics, Stick diagrams, Euler path layout rules, Dynamic CMOS circuits, and Verilog HDL RTL modeling.

Unit 1: MOS Transistor Theory

NMOS and PMOS enhancement mode transistors, Threshold voltage equation, Linear and Saturation region currents, Channel length modulation, Body effect.

Unit 2: CMOS Fabrication & Processing

N-well, P-well, Twin-tub CMOS fabrication process steps, Photolithography, Ion implantation, Latch-up prevention techniques in CMOS.

Unit 3: Circuit Design Rules & Layout

Lambda (λ) and Micron-based design rules, Contact cuts, Transistor sizing, Stick diagrams, Euler path layout optimization, and Interconnect parasitic estimation.

Unit 4: Delay Modeling & Logic Families

RC delay models, Elmore delay calculation, Logical Effort theory, Dynamic CMOS logic, Domino logic, Transmission gates, and Pass transistor logic.

Unit 5: Hardware Description Language (Verilog)

Dataflow, Behavioral, and Structural modeling in Verilog HDL, Testbench design, Synthesis of combinational and sequential circuits (ALU, Shift registers, FSMs).